Impedance Control of a 2-DOF Robot Arm

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I. Introduction & Motivation

In this project, we study this problem on the 2-DOF robot arm shown in Figure 1. Its equations of motion contain nonlinear terms (sines and cosines) and make it a simple but representative manipulator.

Our primary goal is to design and analyze a torque-based *impedance controller* [1], which makes the arm behave like a virtual spring-damper about a desired trajectory. We will (i) derive and linearize the open-loop dynamics, (ii) show how the impedance law changes the local dynamics and stability, and (iii) evaluate the controller in simulation under reference changes and external disturbance torques. Throughout, we carry one concrete example arm so that every matrix below has numbers we can check.

II. System, Inputs & Notation

A. Example arm

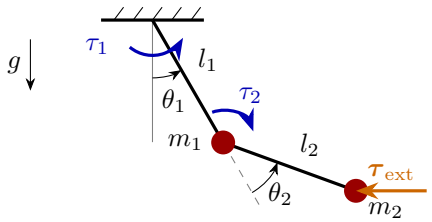


Fig. 1. 2-DOF arm (adapted from [2]). Joint angles θ_1, θ_2 are zero when the arm hangs straight down; θ_2 is relative to link 1. The controller commands the motor torques τ_1 (shoulder) and τ_2 (elbow) in blue. The environment acts through the external torque $\boldsymbol{\tau}_{\text{ext}}$ (orange), which we cannot command.

For all numerical examples we use point masses at the link tips with $m_1 = m_2 = 1$ kg, $l_1 = l_2 = 1$ m, and $g = 9.81$ m/s^2.

B. How we interact with the arm

The arm has two actuators, one at each joint. Our *control input* is the vector of motor torques

$$\mathbf{u} = [\tau_1 \;\; \tau_2]^T \in \mathbb{R}^2,$$

where τ_1 acts at the shoulder (on link 1) and τ_2 at the elbow (between link 1 and link 2). Because each joint has its own motor, both degrees of freedom are directly actuated. The environment (contact, or a push on the arm) acts through the *external torque* $\boldsymbol{\tau}_{\text{ext}} \in \mathbb{R}^2$. It is a disturbance: we do not choose it, and it is zero in free motion. We assume we measure the joint angles and velocities, so the controller sees the full state. In the impedance controller of Section IV, the only other signal we supply is the desired joint configuration $\boldsymbol{\theta}_d$, which sets the equilibrium the virtual spring pulls toward.

C. Notation

Table I lists every symbol used in this proposal. Vectors and matrices are bold; a dot is a time derivative.

TABLE I

Symbols and example values.

Symbol	Meaning	Example
θ_1, θ_2	joint angles (shoulder, elbow); $\theta = 0$ is hanging down	rad
$\dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}}$	joint velocity, acceleration	rad/s, rad/s^2
$m_i,\ l_i$	mass and length of link i	1 kg, 1 m
g	gravitational acceleration	9.81 m/s^2
$\mathbf{M}(\boldsymbol{\theta})$	inertia matrix	2×2
$\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$	Coriolis/centrifugal matrix	2×2
$\boldsymbol{\tau}_g(\boldsymbol{\theta})$	gravity torque vector	N·m
$\mathbf{u} = [\tau_1 \;\; \tau_2]^T$	motor torques (control input)	N·m
$\mathbf{B}$	maps $\mathbf{u}$ to joint torques; $\mathbf{I}_2$ here	2×2
$\boldsymbol{\tau}_{\text{ext}}$	external (contact) torque, uncommanded	N·m
$\mathbf{x}$	state $[\boldsymbol{\theta}^T \;\; \dot{\boldsymbol{\theta}}^T]^T$	$\mathbb{R}^4$
$\mathbf{f}(\mathbf{x}, \mathbf{u})$	state-space dynamics, $\dot{\mathbf{x}} = \mathbf{f}$	
$(\mathbf{x}^*, \mathbf{u}^*)$	equilibrium state–input pair	$(\mathbf{0}, \mathbf{0})$
$\mathbf{A}, \mathbf{B}_{\text{lin}}$	Jacobians of $\mathbf{f}$ at $(\mathbf{x}^*, \mathbf{u}^*)$	4×4, 4×2
$\boldsymbol{\theta}_d,\ \mathbf{e}$	desired angles; error $\boldsymbol{\theta} - \boldsymbol{\theta}_d$	rad
$\mathbf{K}, \mathbf{D}$	virtual stiffness and damping gains	2×2

D. Control objective

Given a constant desired configuration $\boldsymbol{\theta}_d$, design a state-feedback law $\mathbf{u} = \boldsymbol{\pi}(\mathbf{x}, \boldsymbol{\theta}_d)$ such that (i) with $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$ the error $\mathbf{e} = \boldsymbol{\theta} - \boldsymbol{\theta}_d$ converges to zero, and (ii) when $\boldsymbol{\tau}_{\text{ext}} \neq \mathbf{0}$ the arm yields like a spring-damper, $\mathbf{K}\mathbf{e} = \boldsymbol{\tau}_{\text{ext}}$ at rest, with $\mathbf{K}$ and $\mathbf{D}$ chosen by us.

III. Mathematical Model & Linearization

A. Dynamics

As given by [2], the equations of motion (EOM) for a general manipulator with joint angles $\boldsymbol{\theta} \in \mathbb{R}^n$ are

$$\boxed{\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} = \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} + \boldsymbol{\tau}_{\text{ext}}.} \qquad (1)$$

Compared with [2], the term $\boldsymbol{\tau}_{\text{ext}}$ is added so that interaction with the environment can be studied. For our arm ($n = 2$), with $c_2 = \cos\theta_2$, $s_1 = \sin\theta_1$, $s_2 = \sin\theta_2$, $s_{12} = \sin(\theta_1 + \theta_2)$,

$$\mathbf{M}(\boldsymbol{\theta}) = \begin{bmatrix} (m_1 + m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 c_2 & m_2 l_2^2 + m_2 l_1 l_2 c_2 \\ m_2 l_2^2 + m_2 l_1 l_2 c_2 & m_2 l_2^2 \end{bmatrix}$$

$$\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) = \begin{bmatrix} 0 & -m_2 l_1 l_2 (2\dot{\theta}_1 + \dot{\theta}_2) s_2 \\ \frac{1}{2} m_2 l_1 l_2 (2\dot{\theta}_1 + \dot{\theta}_2) s_2 & -\frac{1}{2} m_2 l_1 l_2 \dot{\theta}_1 s_2 \end{bmatrix}$$

$$\boldsymbol{\tau}_g(\boldsymbol{\theta}) = -g \begin{bmatrix} (m_1+m_2)l_1 s_1 + m_2 l_2 s_{12} \\ m_2 l_2 s_{12} \end{bmatrix}, \qquad \mathbf{B} = \mathbf{I}_2,$$

so both joints are directly actuated. With the example values ($m_i = l_i = 1$) these become

$$\mathbf{M} = \begin{bmatrix} 3+2c_2 & 1+c_2 \\ 1+c_2 & 1 \end{bmatrix}, \quad \boldsymbol{\tau}_g = -g \begin{bmatrix} 2s_1 + s_{12} \\ s_{12} \end{bmatrix}.$$

Remark 1 *The matrix* $\mathbf{C}$ *above reproduces the correct Coriolis force* $\mathbf{C}\dot{\boldsymbol{\theta}}$*, but it is not the Christoffel-symbol factorization, so* $\dot{\mathbf{M}} - 2\mathbf{C}$ *is not skew-symmetric. This does not affect the equations of motion, the linearization, or the simulations below. A passivity or Lyapunov argument that relies on skew-symmetry must instead use* $\mathbf{C}_{\text{ch}} = \begin{bmatrix} -h\dot{\theta}_2 & -h(\dot{\theta}_1+\dot{\theta}_2) \\ h\dot{\theta}_1 & 0 \end{bmatrix}$ *with* $h = m_2 l_1 l_2 s_2$*, which gives the same* $\mathbf{C}_{\text{ch}}\dot{\boldsymbol{\theta}}$ *and satisfies* $\dot{\mathbf{M}} - 2\mathbf{C}_{\text{ch}}$ *skew-symmetric. We verified both facts numerically.*

Second-order form. For strictly positive masses and link lengths, $\mathbf{M}(\boldsymbol{\theta})$ is positive definite and therefore invertible, so the EOM solve explicitly for the joint accelerations:

$$\ddot{\boldsymbol{\theta}} = \mathbf{M}(\boldsymbol{\theta})^{-1} \left[-\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} + \boldsymbol{\tau}_{\text{ext}} \right]. \quad (2)$$

State-space form. Stacking $\mathbf{x} = [\theta_1 \;\; \theta_2 \;\; \dot{\theta}_1 \;\; \dot{\theta}_2]^T = [\boldsymbol{\theta}^T \;\; \dot{\boldsymbol{\theta}}^T]^T$ and setting $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$, the dynamics $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ are, explicitly,

$$\mathbf{f}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \dot{\boldsymbol{\theta}} \\ \mathbf{M}(\boldsymbol{\theta})^{-1} \left[-\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{B}\mathbf{u} \right] \end{bmatrix}. \quad (3)$$

The measured output is the pair of joint angles, $\mathbf{y} = \mathbf{C}_y \mathbf{x}$ with $\mathbf{C}_y = [\mathbf{I}_2 \;\; \mathbf{0}_{2\times 2}]$.

B. Open-Loop Linearization

The arm hanging straight down at rest with zero torque is an equilibrium *state–input pair* $(\mathbf{x}^*, \mathbf{u}^*) = (\mathbf{0}, \mathbf{0})$: the velocities are zero and $\boldsymbol{\tau}_g(\mathbf{0}) = \mathbf{0}$, so $\mathbf{f}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$. With $\delta\mathbf{x} = \mathbf{x} - \mathbf{x}^*$ and $\delta\mathbf{u} = \mathbf{u} - \mathbf{u}^*$, the first-order Taylor approximation is

$$\boxed{\delta\dot{\mathbf{x}} = \mathbf{A}\delta\mathbf{x} + \mathbf{B}_{\text{lin}}\delta\mathbf{u},} \quad (4)$$

$$\mathbf{A} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{(\mathbf{x}^*, \mathbf{u}^*)}, \qquad \mathbf{B}_{\text{lin}} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right|_{(\mathbf{x}^*, \mathbf{u}^*)}.$$

Write the lower block of (3) as $\mathbf{M}^{-1}\mathbf{h}$ with $\mathbf{h} = -\mathbf{C}\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_g + \mathbf{B}\mathbf{u}$. Then $\partial(\mathbf{M}^{-1}\mathbf{h})/\partial\boldsymbol{\theta} = (\partial\mathbf{M}^{-1}/\partial\boldsymbol{\theta})\,\mathbf{h} + \mathbf{M}^{-1}\,\partial\mathbf{h}/\partial\boldsymbol{\theta}$, and the first term vanishes because $\mathbf{h}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$. The Coriolis term $\mathbf{C}\dot{\boldsymbol{\theta}}$ is a sum of products $\dot{\theta}_i\dot{\theta}_j$ with coefficients depending on θ_2. Its derivative with respect to $\boldsymbol{\theta}$ is therefore proportional to $\dot{\theta}_i\dot{\theta}_j$, and its derivative with respect to $\dot{\boldsymbol{\theta}}$ is linear in $\dot{\boldsymbol{\theta}}$; both are zero at $\dot{\boldsymbol{\theta}} = \mathbf{0}$. Only gravity contributes, giving

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_2 \\ \mathbf{K}_g & \mathbf{0} \end{bmatrix}, \quad \mathbf{B}_{\text{lin}} = \begin{bmatrix} \mathbf{0} \\ \mathbf{M}_0^{-1}\mathbf{B} \end{bmatrix}, \quad \mathbf{K}_g = \mathbf{M}_0^{-1}\mathbf{G}, \quad (5)$$

where $\mathbf{M}_0 = \mathbf{M}(\mathbf{0})$ and $\mathbf{G} = \partial\boldsymbol{\tau}_g/\partial\boldsymbol{\theta}|_{\mathbf{0}}$, namely

$$\mathbf{M}_0 = \begin{bmatrix} (m_1+m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 & m_2 l_2 (l_1 + l_2) \\ m_2 l_2 (l_1 + l_2) & m_2 l_2^2 \end{bmatrix}, \; \mathbf{G} = -g \begin{bmatrix} (m_1+m_2)l_1 + m_2 l_2 & m_2 l_2 \\ m_2 l_2 & m_2 l_2 \end{bmatrix}.$$

Example values. For $m_i = l_i = 1$ and $g = 9.81$: $\mathbf{M}_0 = \left[\begin{smallmatrix} 5 & 2 \\ 2 & 1 \end{smallmatrix}\right]$, $\mathbf{M}_0^{-1} = \left[\begin{smallmatrix} 1 & -2 \\ -2 & 5 \end{smallmatrix}\right]$, $\mathbf{G} = -g\left[\begin{smallmatrix} 3 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, so

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -9.81 & 9.81 & 0 & 0 \\ 9.81 & -29.43 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{\text{lin}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & -2 \\ -2 & 5 \end{bmatrix}.$$

The matrix $\mathbf{G}$ is symmetric negative definite and $\mathbf{M}_0$ is positive definite, so $\mathbf{K}_g = \mathbf{M}_0^{-1}\mathbf{G}$ is similar to the symmetric negative definite matrix $\mathbf{M}_0^{-1/2}\mathbf{G}\mathbf{M}_0^{-1/2}$ and has real negative eigenvalues $-\mu_i$. Each eigenvalue λ of $\mathbf{A}$ then satisfies $\lambda^2 = -\mu_i$, i.e. $\lambda = \pm j\sqrt{\mu_i}$. For the example, $\mu = g(2 \pm \sqrt{2})$ and the eigenvalues are $\pm j2.40$ and $\pm j5.79$ rad/s, which we confirmed numerically.

C. Open-Loop Analysis

Stability. All four eigenvalues of $\mathbf{A}$ lie on the imaginary axis, so the linearization is marginally stable: it oscillates at 2.40 and 5.79 rad/s and never settles, because the model has no dissipation. Such eigenvalues do not by themselves decide stability of the nonlinear system, but here energy conservation does: with $\mathbf{u} = \mathbf{0}$ the total energy is constant (drift below 10^{-8} J over 10 s in our simulation), so the arm swings indefinitely. This motivates a feedback law that adds damping.

Controllability. The controllability matrix $[\mathbf{B}_{\text{lin}} \;\; \mathbf{A}\mathbf{B}_{\text{lin}} \;\; \mathbf{A}^2\mathbf{B}_{\text{lin}} \;\; \mathbf{A}^3\mathbf{B}_{\text{lin}}] \in \mathbb{R}^{4\times 8}$ has rank 4. This also follows from structure: since $\mathbf{A}\mathbf{B}_{\text{lin}} = \begin{bmatrix} \mathbf{M}_0^{-1} \\ \mathbf{0} \end{bmatrix}$ and $\mathbf{M}_0$ is invertible, the first two blocks $[\mathbf{B}_{\text{lin}} \;\; \mathbf{A}\mathbf{B}_{\text{lin}}]$ alone have rank 4. With one motor per joint, every direction in $(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$ near the hanging equilibrium can be reached.

Observability. With $\mathbf{C}_y = [\mathbf{I}_2 \;\; \mathbf{0}]$, we have $\mathbf{C}_y\mathbf{A} = [\mathbf{0} \;\; \mathbf{I}_2]$, so the first two block rows of the observability matrix already form $\mathbf{I}_4$ and the rank is 4. The velocities can be reconstructed from measured angles, so an observer could replace velocity sensors. We assume full-state measurement and do not need one here.

D. Stabilizability and Detectability

By the PBH test, $(\mathbf{A}, \mathbf{B})$ is *stabilizable* iff $\text{rank}[\lambda\mathbf{I} - \mathbf{A} \;\; \mathbf{B}] = 4$ for every eigenvalue with $\text{Re}\,\lambda \geq 0$, and $(\mathbf{C}, \mathbf{A})$ is *detectable* iff the dual condition holds. Every open-loop eigenvalue of the hanging arm lies on the imaginary axis, so there are no asymptotically stable modes, and for this arm stabilizable is equivalent to controllable and detectable is equivalent to observable: losing any mode is fatal. Table II summarizes the test at all four equilibria $\boldsymbol{\theta}^* \in \{0, \pi\}^2$ and with actuators or sensors removed, computed from numerical Jacobians of the nonlinear model.

TABLE II

STRUCTURAL PROPERTIES OF THE LINEARIZATION (RANK OUT OF 4).

Case	Inputs / outputs	Ctrb./Obsv.	Stab./Det.
All 4 equilibria, $g = 9.81$	$\mathbf{u} = [\tau_1, \tau_2]$, $\mathbf{y} = \boldsymbol{\theta}$	4 / 4	yes / yes
All 4 equilibria, $g = 9.81$	τ_1 only or τ_2 only	4	yes
Hanging, $g = 9.81$	any single θ_i or $\dot{\theta}_i$	4	yes
Hanging, $g = 0$ (horizontal)	τ_1 only or τ_2 only	2	**no**
Hanging, $g = 0$ (horizontal)	$\mathbf{y} = \dot{\boldsymbol{\theta}}$, θ_1, or θ_2 only	2	**no**

Unstable equilibria. The upright configurations have eigenvalues in the open right half-plane (e.g. $\pm 2.40, \pm 5.79$ for $\boldsymbol{\theta}^* = (\pi, 0)$), but they are equally controllable and observable, so each can be stabilized locally by state feedback. Controllability is a property of $(\mathbf{A}, \mathbf{B})$, not of open-loop stability.

Role of gravity. With a single motor at joint j, $\mathbf{b}_j = [\mathbf{0}; \mathbf{M}_0^{-1}\mathbf{e}_j]$. If $g = 0$, $\mathbf{A}^2 = \mathbf{0}$ and the controllability matrix is $[\mathbf{b}_j \quad \mathbf{A}\mathbf{b}_j]$ with rank 2. Two modes at $s = 0$ are lost, so the arm is not stabilizable. Likewise, velocity sensing alone cannot recover the absolute angles. With $g \neq 0$, $\mathbf{K}_g$ is invertible and couples the joints, so one motor, or one sensor of any kind, recovers rank 4.

When can a single motor lose a mode? Let the modes be $\mathbf{G}\mathbf{v} = -\mu\mathbf{M}_0\mathbf{v}$. The corresponding left eigenvector of $\mathbf{A}$ satisfies $\mathbf{w}^T\mathbf{b}_j \propto v_j$, and with $\mathbf{y} = \theta_j$ the right eigenvector gives $\mathbf{C}\mathbf{x} \propto v_j$. A motor or angle sensor at joint j therefore misses a mode iff that mode shape has no motion at joint j. Imposing $\mathbf{G}\mathbf{e}_1 \parallel \mathbf{M}_0\mathbf{e}_1$ gives $m_1 l_1 l_2 = 0$ (elbow only), and $\mathbf{G}\mathbf{e}_2 \parallel \mathbf{M}_0\mathbf{e}_2$ gives $l_1 = 0$ (shoulder only). For a real arm neither condition holds. As $m_1 \to 0$, however, the elbow-only margin $\min|v_2|/\|\mathbf{v}\|$ shrinks like $m_1/2$. The rigid "swing" mode is then barely excited by an elbow torque. At $m_1 = 0.01$ a numerical Kalman rank test already reports rank 2 although the system is controllable, so near-degenerate cases should be decided by the PBH or modal test.

Constructive stabilization. Linearizing the impedance law (6) at $\boldsymbol{\theta}_d = \mathbf{0}$ gives the state feedback $\delta\mathbf{u} = \mathbf{F}\delta\mathbf{x}$ with $\mathbf{F} = -[\mathbf{G} + \mathbf{K} \quad \mathbf{D}]$, and $\mathbf{A} + \mathbf{B}_{\text{lin}}\mathbf{F} = \mathbf{A}_{cl}$ is Hurwitz (Sec. IV). This exhibits stabilizability directly. In closed loop the disturbance channel $(\mathbf{A}_{cl}, [\mathbf{0}; \mathbf{M}_d^{-1}])$ stays controllable, and for $\mathbf{K} = k\mathbf{I}$, $\mathbf{D} = d\mathbf{I}$ the position block of its controllability Gramian is $\mathbf{I}/(2dk)$, independent of $\mathbf{M}_d$.

IV. IMPEDANCE CONTROL (PRIMARY GOAL)

Let $\boldsymbol{\theta}_d$ be a desired (constant) joint configuration. We apply the torque law

$$\boxed{\mathbf{u} = -\boldsymbol{\tau}_g(\boldsymbol{\theta}) + \mathbf{K}(\boldsymbol{\theta}_d - \boldsymbol{\theta}) - \mathbf{D}\dot{\boldsymbol{\theta}},} \tag{6}$$

where $\mathbf{K}, \mathbf{D} \in \mathbb{R}^{2\times 2}$ are symmetric positive definite stiffness and damping gains. Gravity is compensated, and the remaining terms act as a virtual spring-damper at each joint. Substituting into (1) gives the closed loop

$$\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \Big(\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{D}\Big)\,\dot{\boldsymbol{\theta}} + \mathbf{K}(\boldsymbol{\theta} - \boldsymbol{\theta}_d) = \boldsymbol{\tau}_{\text{ext}}.$$

With error $\mathbf{e} = \boldsymbol{\theta} - \boldsymbol{\theta}_d$ and state $\mathbf{z} = [\mathbf{e}^T \ \dot{\mathbf{e}}^T]^T$, linearizing about $\mathbf{z} = \mathbf{0}$, $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$ yields

$$\dot{\mathbf{z}} = \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}_d^{-1}\mathbf{K} & -\mathbf{M}_d^{-1}\mathbf{D} \end{bmatrix}}_{\mathbf{A}_{cl}} \mathbf{z} + \begin{bmatrix} \mathbf{0} \\ \mathbf{M}_d^{-1} \end{bmatrix} \boldsymbol{\tau}_{\text{ext}}, \tag{7}$$

where $\mathbf{M}_d = \mathbf{M}(\boldsymbol{\theta}_d)$. This is a mass-spring-damper whose stiffness and damping we choose. Because $\mathbf{M}_d$, $\mathbf{K}$, and $\mathbf{D}$ are positive definite, $\mathbf{A}_{cl}$ is Hurwitz. For the example arm with $\boldsymbol{\theta}_d = \mathbf{0}$, $\mathbf{K} = 20\,\mathbf{I}$ and $\mathbf{D} = 5\,\mathbf{I}$, the eigenvalues of $\mathbf{A}_{cl}$ are about -24.4, -4.79, and $-0.43 \pm j1.80$. We will verify this in simulation. To keep the project tractable, we use diagonal gains and simulate in joint space only.

Planned analysis and simulation

- Eigenvalues of $\mathbf{A}_{cl}$ for a few choices of $\mathbf{K}$ and $\mathbf{D}$ at one or two values of $\boldsymbol{\theta}_d$.
- Nonlinear simulation (numerical ODE integration): step in $\boldsymbol{\theta}_d$ and a short disturbance torque pulse $\boldsymbol{\tau}_{\text{ext}}$, compared against the linear model.
- Effect of stiffness versus damping on the response.

V. PLAN & DELIVERABLES

TABLE III

PROPOSED TIMELINE.

Stage	Task
1	Model, linearize, and verify numerically
2	Open-loop eigenvalue, controllability, observability
3	Impedance controller, closed-loop analysis, simulation
4	Final report and presentation

We will implement the nonlinear model and controllers in Python (NumPy/SciPy) and report the open-loop and closed-loop eigenvalues and the step and disturbance responses for a few gain choices. Success means a stable, well-damped closed loop that tracks $\boldsymbol{\theta}_d$ and deflects predictably under $\boldsymbol{\tau}_{\text{ext}}$. All work is in simulation, with no hardware.

REFERENCES

[1] N. Hogan, "Impedance control: An approach to manipulation: Part i—theory," *Journal of Dynamic Systems, Measurement, and Control*, vol. 107, no. 1, pp. 1–7, 03 1985. [Online]. Available: https://doi.org/10.1115/1.3140702

[2] R. Tedrake, *Underactuated Robotics*, 2023. [Online]. Available: https://underactuated.csail.mit.edu